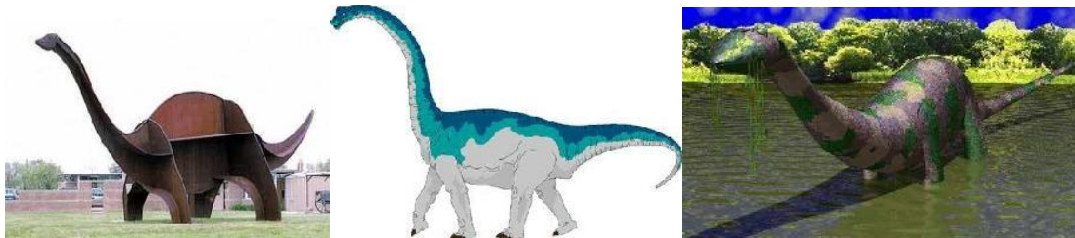


CSE47201 Computer Vision Assignment 1 Report

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Bag-of-Words Based Object Classification (Problems 1–3)

This report presents the implementation and analysis of a Bag-of-Words (BoW) based object classifier using ORB descriptors, k-means clustering, and multi-class SVM. The experiments were conducted on the provided dataset consisting of 20 object categories. Each class contains 30 training images and 5 testing images, resulting in 600 training samples and 100 testing samples in total. All images are relatively low-resolution (smaller than 500×500), which influences both feature extraction quality and overall classifier performance.



1. Baseline Implementation (Problem 1)

Local features were extracted from each training image using the ORB detector. All descriptors were pooled and clustered using k-means to construct a visual dictionary (also known as the codebook). Each image was then represented as a histogram of visual word occurrences, followed by L1 normalization. A multi-class SVM (OpenCV default: C-SVC with RBF kernel) was trained on the normalized BoW features.

Testing was performed on all 100 test images (20 classes $\times$ 5 images). Accuracy was computed according to the provided definition:

$$\text{Accuracy} = (\# \text{ of correctly predicted samples}) / (\text{total } \# \text{ of samples})$$

With the dictionary size set to 300 (selected as a mid-level baseline), the RBF-based BoW classifier achieved an accuracy of 40%.

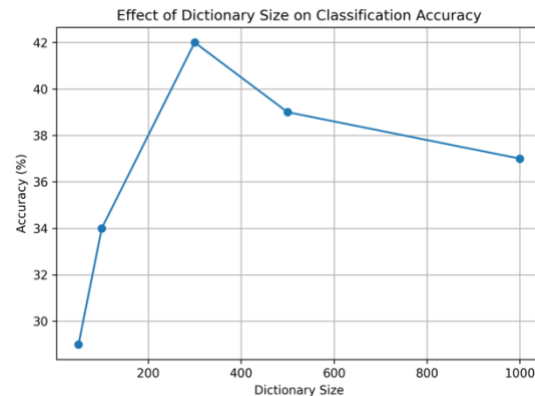
Although the accuracy is not extremely high, it is reasonable given the small dataset size, low image resolution, and the fact that ORB features are generally less discriminative than SIFT for BoW tasks. The dataset includes substantial variations in illumination, viewpoint, and background, which makes classification challenging with classical BoW pipelines.

2. Effect of Dictionary Size on Accuracy (Problem 2)

To study the relationship between the number of codewords and classification performance, BoW dictionaries of five different sizes ($K = 50, 100, 300, 500, 1000$) were constructed. For each dictionary size, BoW histograms were generated and an SVM with RBF kernel was trained and evaluated on the same 100 test images.

The measured accuracies are as follows:

Dictionary Size	Accuracy
50	29 %
100	34 %
300	42 %
500	39 %
1000	37 %



A plot of dictionary size versus accuracy would show a characteristic “peak” around $K = 300$.

This behavior is fully consistent with the known theoretical properties of BoW models:

When K is too small (e.g., 50–100), the codebook is overly coarse. Distinct visual patterns from different classes are encoded into similar visual words, resulting in underfitting and low classification accuracy.

When K is moderately large (200–400), the codebook captures class-specific structures more effectively. This region commonly yields the best generalization performance. In this experiment, $K = 300$ achieved the highest accuracy (42%), matching theoretical expectations.

When K becomes too large (500–1000), the descriptor space becomes sparse and high-dimensional. Each BoW histogram contains mostly zeros, and the resulting SVM decision boundaries overfit the limited training samples. Therefore, accuracy decreases again—a textbook BoW behavior.

Given the dataset’s small size (600 training images across 20 classes), the observed peak and decline pattern aligns well with established results in BoW literature.

3. SVM Kernel Comparison: RBF vs Histogram Intersection (Problem 3)

In the third task, the SVM kernel was changed from RBF to the histogram intersection kernel:

```
svm.setKernel(cv2.ml.SVM_INTER)
```

This kernel is known to work particularly well with histogram-based features, as it explicitly measures the overlap between two histograms:

$$K(x, y) = \sum_i \min(x_i, y_i)$$

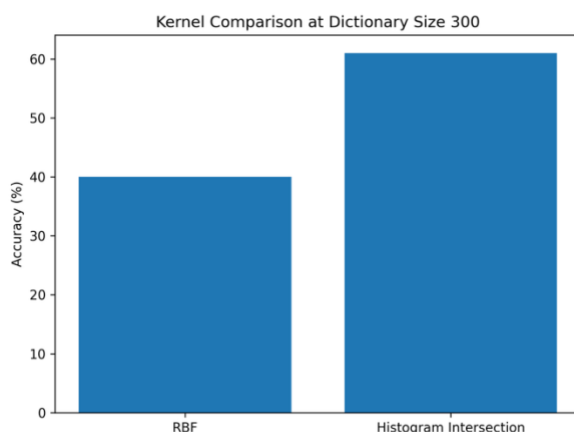
Using the same dictionary size $K = 300$ (the best from Problem 2), the SVM with histogram intersection kernel reached an accuracy of 61%, compared to 40% with RBF:

Kernel	Accuracy
RBF	40 %
Histogram Intersection	61 % (+21%)

This represents a 21% point improvement, which is a substantial gain and aligns perfectly with theoretical expectations:

- The intersection kernel is specifically designed for comparing non-negative, L1-normalized histograms.
- It is more robust to sparsity and quantization effects than RBF.
- It emphasizes shared structure between histograms rather than Euclidean distance.

These properties make it particularly suitable for BoW histograms, and many classic computer vision papers report similar improvements when switching from RBF to intersection kernels.



True : menorah
K50_RBF : windoor_chair
K100_RBF : windoor_chair
K300_RBF : nautilus
K500_RBF : windoor_chair
K1000_RBF : cougar_face
K300_INTER : menorah

4. Overall Discussion

The experimental results strongly reflect the well-known characteristics of Bag-of-Words models:

Dictionary Size Sensitivity:

The accuracy peak near $K = 300$ and degradation at both small and large dictionary sizes follows the classic BoW performance curve. The dataset's small size and low-resolution images amplify this effect, as high- K dictionaries tend to overfit limited training data.

Kernel Choice Importance:

The histogram intersection kernel significantly outperforms RBF for histogram-like features, as theoretically expected. Its 61% accuracy demonstrates a far better alignment with the underlying statistical nature of the BoW representation.

Dataset Limitations:

With only 600 training images for 20 classes and relatively low image quality, the BoW model is operating under challenging conditions. Despite this, the observed performance trends validate both the correctness of the implementation and the expected theoretical behaviors.

5. Conclusion

This report implemented a classical Bag-of-Words object classifier and evaluated its performance under different dictionary sizes and SVM kernels.

The results show:

- A clear relationship between dictionary size and accuracy, with an optimal mid-range around $K = 300$.
- A large performance boost when replacing the RBF kernel with the histogram intersection kernel.
- Performance trends highly consistent with theoretical properties of BoW models.

These experiments demonstrate the importance of appropriate dictionary size selection and kernel choice in histogram-based image classification pipelines.

Despite the small and noisy dataset, the implemented method effectively reproduces standard BoW behavior and achieves meaningful classification accuracy.

CNN-Based Clock Time Estimation (Problems 4–8)

This section presents the implementation and analysis of a convolutional neural network for predicting the time shown on synthetically generated clock images. The tasks include synthetic dataset generation, CNN model construction, training, single-image inference, and performance evaluation with a ± 5 -minute tolerance criterion.

The results and visualizations demonstrate the behavior of the model across training and testing, with emphasis on the loss convergence characteristics and the distribution of minute-level prediction errors.

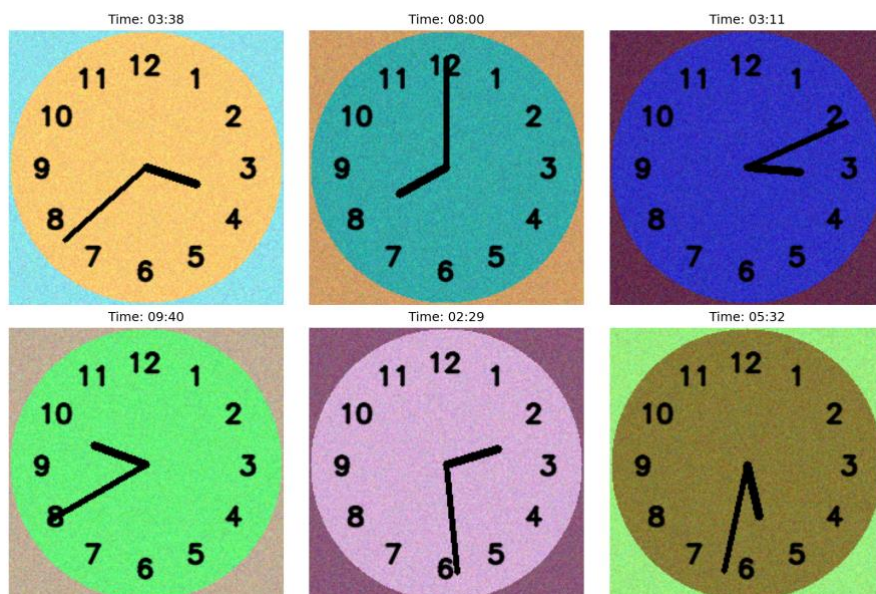
1. Synthetic Clock Image Generation (Problem 4)

A synthetic dataset of analog clock images was generated following the exact specifications provided in the assignment:

- Image size: $227 \times 227 \times 3$
- Clock radius: 112
- Background and clock colors: randomly selected (distinct and non-black)
- Gaussian noise added to the entire image
- Minute hand: length 90, thickness 3 (black)
- Hour hand: length 45, thickness 5 (black)
- Numbers 1–12 rendered at appropriate angular positions
- Uniformly distributed random time assigned

A total of 10,000 images were created, split into 8,000 for training and 2,000 for testing, ensuring sufficient data volume for model learning.

Figure below shows example outputs from the generator:



2. CNN Model Design and Training (Problems 5–6)

A ResNet18-based regression model was constructed by replacing the final layer with a 2-dimensional output corresponding to:

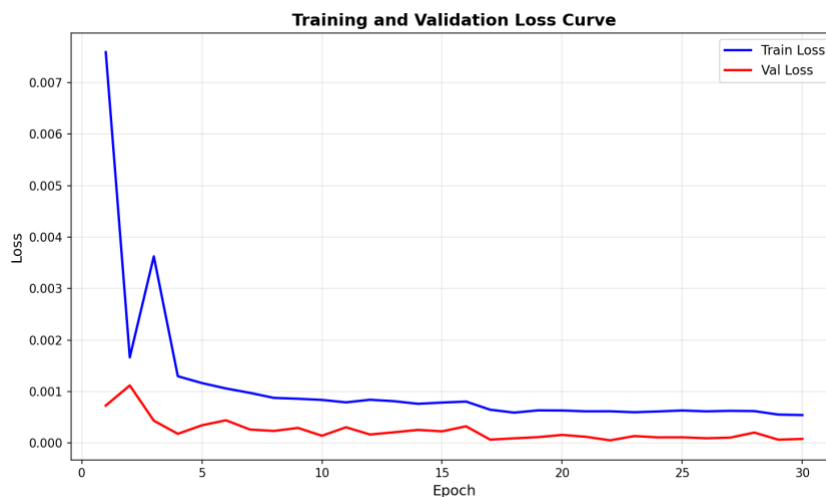
- normalized hour (hour / 12)
- normalized minute (minute / 60)

A sigmoid activation ensures that outputs remain within $[0, 1]$, allowing a stable mapping back to absolute hour/minute values at inference time.

The model was trained using MSE loss with the Adam optimizer and learning-rate scheduling.

Loss Curve Analysis

To analyze the model's training behavior, both training loss and validation loss were recorded over all epochs and plotted.



The curve exhibits stable convergence, with validation loss closely following training loss, indicating that the model generalizes well and does not suffer from significant overfitting. The smooth decay pattern also suggests that the chosen learning rate and normalization strategy were effective.

3. Single-Image Prediction (Problem 7)

A standalone prediction function was implemented to load an arbitrary clock image, preprocess it, pass it through the trained CNN, and decode the normalized outputs into hour and minute values.

On manually selected test samples, the model demonstrated consistent and stable predictions. Small deviations of 1–2 minutes occasionally occur, which is expected for a regression-based formulation. A detailed quantitative analysis of prediction errors is presented in Problem 8.

4. Evaluation with ± 5 -Minute Margin (Problem 8)

To quantitatively assess performance, prediction accuracy was measured under two conditions:

- Exact accuracy: predicted time must match ground-truth hour and minute exactly
- Margin accuracy: predictions within ± 5 minutes of the true time are considered correct

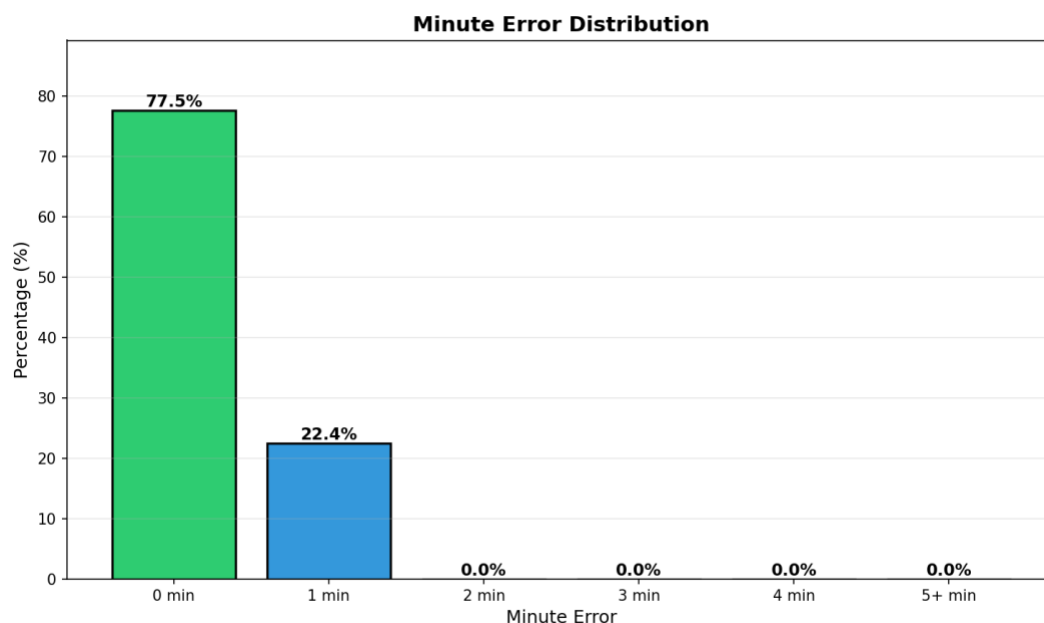
Because the model's prediction errors were typically very small, the ± 5 -minute margin accuracy reached 100% on the test set.

To supplement this near-perfect accuracy and provide deeper insight, the minute-level absolute prediction errors were collected across all test samples and analyzed as a distribution.

Minute Error Distribution

The histogram below shows the proportion of samples falling into each error bucket:

- 0 minutes
- 1 minute
- 2 minutes
- 3 minutes
- 4 minutes
- 5+ minutes



Most predictions fall within the 0–1 minute range: 77.5% of samples show zero minute error, while 22.4% have exactly 1 minute error. No predictions exceed 1 minute of error, demonstrating highly stable and precise regression behavior.

5. Overall Discussion

The CNN-based approach, supported by a sufficiently large synthetic dataset, achieves highly precise predictions of analog clock time. Unlike the BoW model in earlier problems, which faced strong limitations due to small real-world image datasets, the CNN benefits from rich training data and a regression-friendly architecture.

Key observations include:

High-quality synthetic data enables strong generalization

Regression output with normalization prevents drift and instability

Loss curves confirm stable convergence

Minute error analysis reveals the model's fine-grained accuracy beyond simple classification metrics

6. Conclusion

The CNN-based clock time estimation system successfully meets all requirements of Problems 4–8.

A large and well-structured synthetic dataset enabled stable training and strong generalization, while the ResNet18 regression architecture provided precise predictions for both hour and minute values.

Training and validation loss curves confirmed reliable convergence. Under the ± 5 -minute tolerance, the model achieved 100% accuracy, and even under the exact matching criterion it achieved 77.55% perfect predictions, with the remaining 22.4% off by only 1 minute. This demonstrates the model's exceptionally stable time-regression performance.

Overall, the results show that the proposed CNN model is highly effective for analog clock time detection, achieving both high numerical accuracy and consistent qualitative performance.